

2015 HSC Mathematics General 2 Marking Guidelines

IMPORTANT:

This resource was developed to support a previous version of the syllabus.
It may contain content that differs from the current syllabus.

Section I

Multiple-choice Answer Key

Question	Answer
1	D
2	A
3	B
4	A
5	B
6	C
7	B
8	C
9	B
10	A
11	D
12	C
13	A
14	C
15	B
16	D
17	C
18	A
19	D
20	A
21	C
22	D
23	D
24	B
25	C

Section II

Question 26 (a)

Criteria	Marks
• Provides the correct answer	2
• Makes progress towards the correct answer	1

Sample answer:

$$\frac{18 \times 26}{4} = 117$$

Question 26 (b)

Criteria	Marks
• Provides the correct answer	2
• Makes progress towards the correct answer	1

Sample answer:

$$\begin{aligned} \text{Dosage} &= \frac{35 \times 3150}{70} \\ &= 1575 \text{ mg} \end{aligned}$$

$$\begin{aligned} \text{Number of tablets} &= \frac{1575}{525} \\ &= 3 \end{aligned}$$

Question 26 (c)

Criteria	Marks
• Provides the correct answer	2
• Makes progress towards the correct answer	1

Sample answer:

$$\begin{aligned} \frac{40}{360} \times 2 \times \pi \times 6400 &= 4468.042\dots \\ \text{distance} &= 4468 \text{ km} \end{aligned}$$

Question 26 (d)

Criteria	Marks
• Provides the correct answer	2
• Makes progress towards the correct answer	1

Sample answer:

$$A = 320 (1 + 0.029)^5$$

amount = \$369.17

Question 26 (e) (i)

Criteria	Marks
• Provides the correct answer	1

Sample answer:

$$1 - 0.83 = 0.17$$

Question 26 (e) (ii)

Criteria	Marks
• Provides the correct answer	1

Sample answer:

$$1 - 0.14 = 0.86$$

Question 26 (f)

Criteria	Marks
• Provides a correct numerical expression for the area	2
• Makes progress towards the correct solution	1

Sample answer:

$$0.71 \times 4 \times \pi \times 6400^2 = 365\,450\,163.7\dots$$

$$\text{area} = 3.7 \times 10^8 \text{ km}^2$$

Question 26 (g)

Criteria	Marks
• Provides a correct numerical expression of the total	3
• Makes significant progress towards the correct solution	2
• Correctly calculates the cost of the calls or makes progress towards the cost of the data	1

Sample answer:

$$\$561 - \$550 = \$11$$

$$\begin{aligned} 1.7 \text{ GB} &= 1.7 \times 1024 \text{ MB} \\ &= 1740.8 \text{ MB} \end{aligned}$$

$$(1740.8 - 500) \times \$0.0293 = \$36.36$$

$$\begin{aligned} \therefore \text{Last month's bill} &= \$49 + \$11 + \$36.36 \\ &= \$96.36 \end{aligned}$$

Question 27 (a)

Criteria	Marks
• Provides the correct solution	2
• Makes progress towards the correct solution	1

Sample answer:

$$\frac{\text{shadow}}{19.2} = \frac{5}{1.65}$$

$$\begin{aligned} \text{shadow} &= \frac{5}{1.65} \times 19.2 \\ &= 58.18... \\ &= 58 \text{ m} \end{aligned}$$

Question 27 (b)

Criteria	Marks
• Provides a correct numerical expression for the number of drops per minute	2
• Makes progress towards the correct solution	1

Sample answer:

$$\frac{2400 \times 15}{12 \times 60} = 50 \text{ drops/min}$$

Question 27 (c) (i)

Criteria	Marks
• Provides the correct values for x and y	2
• Provides one correct value, or equivalent merit	1

Sample answer:

$$x + 18y = 1510$$

$$x + 36y = 2770$$

$$18y = 1260$$

$$y = 70$$

$$x + 18 \times 70 = 1510$$

$$x = 1510 - 18 \times 70$$

$$= 250$$

Question 27 (c) (ii)

Criteria	Marks
• Provides the correct answer	2
• Attempts to use \$4800 and the solution from (i) to find the number of months or equivalent merit	1

Sample answer:

$$(\$4800 - 250) \div 70 = 65 \text{ months}$$

Question 27 (d) (i)

Criteria	Marks
• Determines that it is an outlier, justified with correct calculations	3
• Makes significant progress towards the correct solution	2
• Calculates correctly the IQR, or equivalent merit	1

Sample answer:

\$300 \$490 \$520 \$590 \$660 \$680 \$970
 Q_L Median Q_U

$$\begin{aligned}\text{IQR} &= 680 - 490 \\ &= \$190\end{aligned}$$

$$\$680 + 1.5 \times \$190 = \$965$$

$\therefore$ Since $\$970 > \965 , it is an outlier.

Question 27 (d) (ii)

Criteria	Marks
• Provides the correct answer	1

Sample answer:

The standard deviation is not affected.

Question 27 (e)

Criteria	Marks
• Provides the correct answer	3
• Makes significant progress towards the correct answer	2
• Provides one correct conversion	1

Sample answer:

$$\begin{aligned}42 \text{ MB} &= 42 \times 2^{20} \times 8 \\ &= 352\,321\,536 \text{ bits} \\ \text{Time} &= \frac{352\,321\,536}{500 \times 1000} \\ &= 704.643 \dots \text{ seconds} \\ &= 11.744 \dots \text{ minutes} \\ &= 11 \text{ minutes } 45 \text{ seconds}\end{aligned}$$

Question 28 (a)

Criteria	Marks
Provides a correct numerical expression for the area	1

Sample answer:

$$\pi(5^2 - 3^2) = 50.265\dots$$

$$\text{area} \doteq 50.3 \text{ cm}^2$$

Question 28 (b)

Criteria	Marks
Determines that the claim is correct, justified with correct and appropriate calculations	2
Correctly calculates one z-score or equivalent merit	1

Sample answer:

Maths z-score: $\frac{74 - 70}{6.5} = 0.6153$

English z-score: $\frac{80 - 75}{8} = 0.625$

$\therefore$ Kristoff is correct since English z-score is higher than Maths z-score.

Question 28 (c)

Criteria	Marks
• Provides the correct answer	3
• Makes significant progress towards the correct answer	2
• Makes some relevant progress, eg correct value of h or correct conversion to litres	1

Sample answer:

Simpson's Rule:

$$V = \frac{h}{3} [A_L + 4A_M + A_R]$$

$$= \frac{15}{3} [45 + 4 \times 180 + 35]$$

$$\text{Volume} = 4000 \text{ cm}^3$$

$$\text{Capacity} = 4000 \text{ mL}$$

$$= 4 \text{ L}$$

Question 28 (d)

Criteria	Marks
• Provides the correct answer	2
• Substitutes correctly into the formula	1

Sample answer:

$$C = \frac{5}{9}(F - 32)$$

$$3 = \frac{5}{9}(F - 32)$$

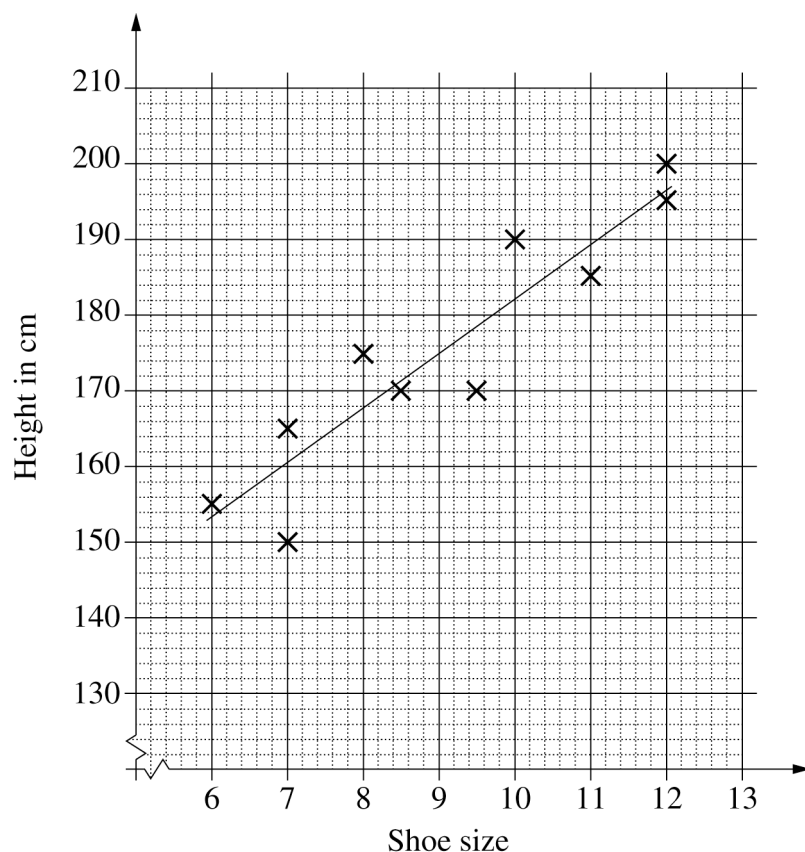
$$\frac{27}{5} = F - 32$$

$$F = 32 + \frac{27}{5}$$

$$\therefore 3^\circ\text{C} = 37.4^\circ\text{F}$$

Question 28 (e) (i)

Criteria	Marks
• Correctly completes the scatterplot AND draws line of fit	2
• Plots the points correctly or draws line of fit using plotted points	1

Sample answer:**Question 28 (e) (ii)**

Criteria	Marks
• Provides the correct answer from the line of fit drawn	1

Sample answer:

$$175 \text{ cm} - 164 \text{ cm} = 11 \text{ cm}$$

Question 28 (e) (iii)

Criteria	Marks
• Provides a correct explanation	1

Sample answer:

All points do not lie on a line.

Question 28 (f) (i)

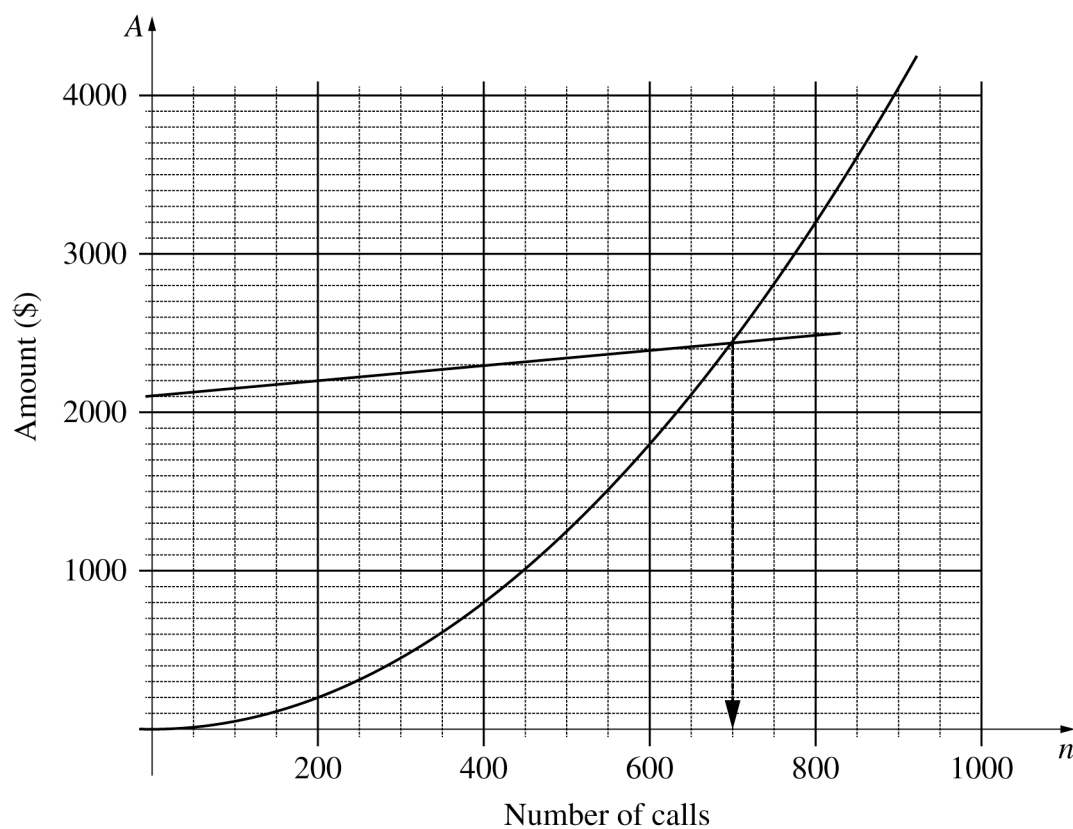
Criteria	Marks
Provides the correct equation	1

Sample answer:

$$C = 2100 + 0.5n$$

Question 28 (f) (ii)

Criteria	Marks
Correctly graphs the equation and determines the correct breakeven point	2
Makes progress towards the correct solution	1

Sample answer:

The charity needs to make 700 calls.

Question 29 (a)

Criteria	Marks
• Provides a correct numerical expression for the amount paid	2
• Makes progress towards the correct solution	1

Sample answer:

$$\left(\frac{18.4}{100} \times 425\right) \times \frac{12}{365} = \$2.57$$

$$\therefore \text{Total amount paid} = \$2.57 + \$425 = \$427.57$$

Question 29 (b)

Criteria	Marks
• Provides the correct answer	2
• Makes progress towards the correct answer	1

Sample answer:

$$(8 \times 12 \times \$1880) - 80\,000$$

$$= \$180\,480 - 80\,000$$

$$= \$100\,480$$

Question 29 (c) (i)

Criteria	Marks
• Determines the scale and correctly verifies the area	2
• Makes progress towards verifying the area, eg finds correct scale	1

Sample answer:

$$1 \text{ cm} = 3 \text{ m} \quad (12 \div 4 = 3)$$

$$6 \text{ cm} \Rightarrow 18 \text{ m}$$

$$\begin{aligned} \text{Area} &= 18 \times 12 \text{ m}^2 \\ &= 216 \text{ m}^2 \end{aligned}$$

Question 29 (c) (ii)

Criteria	Marks
• Provides the correct answer	3
• Makes significant progress towards the correct answer	2
• Calculates the volume of water, or equivalent merit	1

Sample answer:

$$\begin{aligned} V &= 216 \times 0.005 \text{ (m}^3\text{)} \\ &= 1.08 \text{ (m}^3\text{)} \end{aligned}$$

$$\therefore 1.08 = \pi \times 1.8^2 \times h$$

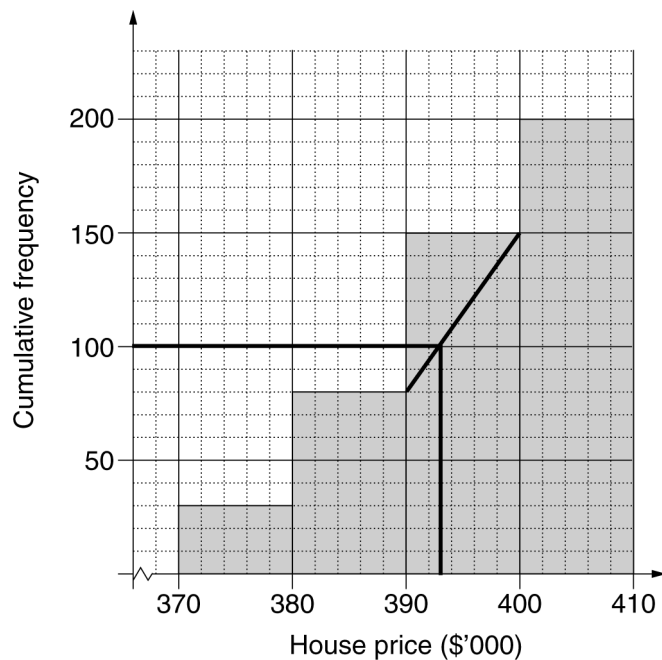
$$h = 0.106103\dots$$

$$= 0.106\dots \text{ m}$$

$$\begin{aligned} \text{Increase in depth} &\doteq 0.106 \text{ m} \\ &\doteq 106 \text{ mm} \end{aligned}$$

Question 29 (d) (i)

Criteria	Marks
• Provides a correct estimate	1

Sample answer:

Median House Price = \$393 000

Question 29 (d) (ii)

Criteria	Marks
• Provides the correct answer	3
• Makes significant progress towards the correct answer	2
• Provides at least 2 correct class centres with correct corresponding frequencies	1

Sample answer:

Class centre (\$'000)	Frequency
375	30
385	50
395	70
405	50

$$\bar{x} = \frac{(375 \times 30) + (385 \times 50) + (395 \times 70) + (405 \times 50)}{200}$$

$$= \$392$$

 $\therefore$ Mean House Price = \$392 000

Question 29 (e)

Criteria	Marks
• Provides a correct explanation	2
• Makes a correct interpretation using the graph	1

Sample answer:

The diving board is 8 m above the water (where the graph cuts the vertical axis). To find how high the diver was above the board, subtract 8 metres from the maximum height of the graph which is approximately 9.2 m (or which occurs at 0.5 on the horizontal axis).

Question 30 (a)

Criteria	Marks
• Provides a correct numerical expression for the saving	2
• Makes progress towards the correct solution	1

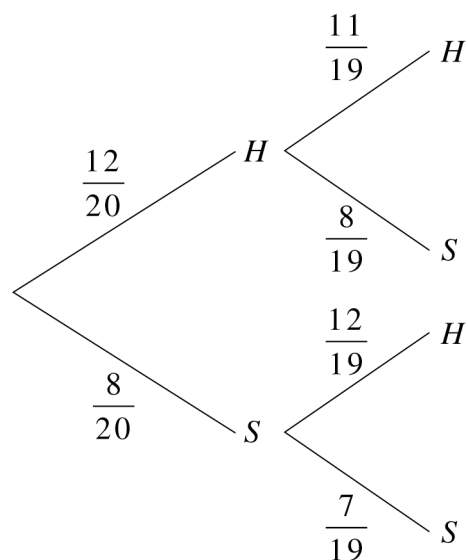
Sample answer:

$$20 \times \frac{21}{1000} \times 0.31 \times 11 \times 7 \times 24 = \$240.61$$

$\therefore$ The school saves \$240.61

Question 30 (b)

Criteria	Marks
• Provides the correct probability	3
• Makes significant progress towards the correct answer	2
• Correctly completes the probability tree, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 P(\text{one of each type}) &= P(HS) + P(SH) \\
 &= \frac{12}{20} \times \frac{8}{19} + \frac{8}{20} \times \frac{12}{19} \\
 &= \frac{48}{95} \\
 &\text{OR } 0.505 \quad \text{or } 50.5\%
 \end{aligned}$$

Question 30 (c) (i)

Criteria	Marks
• Provides a correct numerical expression for the value	1

Sample answer:

$$\begin{aligned}\text{Value} &= \$200 \times 55.68446 \\ &= \$11\,136.89\end{aligned}$$

Question 30 (c) (ii)

Criteria	Marks
• Provides a correct numerical expression for the monthly repayment	2
• Makes progress towards the correct solution	1

Sample answer:

$$\begin{aligned}10.8\% \text{ p.a.} &= (10.8 \div 12)\% \text{ per/month} \\ &= 0.9\% \\ &= 0.009\end{aligned}$$

$$6 \text{ years} = 72 \text{ months}$$

$$\begin{aligned}\text{Monthly repayment} &= \$21\,500 \div 52.82118 \\ &= \$407.03 \\ &= \$407\end{aligned}$$

Question 30 (d)

Criteria	Marks
• Provides a correct numerical expression for the stopping distance	2
• Makes progress towards the correct solution	1

Sample answer:

$$\begin{aligned}\text{Reaction – time distance} &= s \times t \\ &= \frac{110\,000}{3600} \times 2 \\ &= 61.11\ldots\end{aligned}$$

$$\begin{aligned}\text{Stopping distance} &= 61.11\ldots + 59.2 \\ &= 120.311\ldots \\ &= 120 \text{ m}\end{aligned}$$

Question 30 (e) (i)

Criteria	Marks
Correctly substitutes into the cosine rule to give a correct numerical expression for PC	1

Sample answer:

$$PC = \sqrt{5.4^2 + 1.8^2 - 2(5.4)(1.8)\cos 108^\circ}$$

$$= \sqrt{38.407\dots}$$

$$\therefore PC = 6.1973\dots$$

Question 30 (e) (ii)

Criteria	Marks
Provides the correct answer	4
Makes significant progress towards finding PE, or equivalent merit	3
Correctly calculates an angle in $\triangle PSC$, or equivalent merit	2
Makes some progress towards the correct solution, eg attempts to find an angle in $\triangle PSC$ OR writes a correct trig equation involving h	1

Sample answer:

$$\frac{\sin C}{5.4} = \frac{\sin 108^\circ}{6.197}$$

$$\sin C = 0.8287\dots$$

$$\therefore C = 55.969\dots^\circ$$

$$= 55^\circ 58'$$

$$\text{In } \triangle PCE, \cos 55^\circ 58' = \frac{PE}{6.197}$$

$$\therefore PE = 6.197 \times \cos 55^\circ 58'$$

$$= 3.468\dots$$

$$\therefore h = 3.468 - 2.1$$

$$= 1.3680\dots$$

$$\text{height} \doteq 1.37 \text{ m}$$

2015 HSC Mathematics General 2

Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	MMI Sig fig, sci notation	MGP-2
2	1	AM3 Simple Algebra (Adding like terms)	MG2H-3
3	1	FMI Net pay	MGP-6
4	1	DSI Classify Data	MGP-7
5	1	AM5 Identify Non-linear graphs	MG2H-3
6	1	DS4 Interpreting box and whisker plot	MG2H-1, 7
7	1	MM5 Radial Survey	MG2H-4
8	1	MM4 Volume of a cone/pyramid	MG2H-4
9	1	MM5 Right Angled Trigonometry	MG2H-4
10	1	FsDr2 Depreciation	MGP-6
11	1	AM3 Index Laws	MG2H-3
12	1	MM4 Percentage Error	MG2H-5
13	1	AM2 $y = mx + b$	MGP-2
14	1	MM6 Time Differences	MG2H-4
15	1	FM3 Tax	MGP-6
16	1	PB2 Probability	MG2H-8
17	1	FM4 PV of annuity (Compound interest)	MG2H-6
18	1	PB2 Counting Techniques	MG2H-8
19	1	FsHe3 Life Expectancy	MG2H-2
20	1	DS5 Normal Distribution	MG2H-7
21	1	PB2 Counting Techniques	MG2H-8
22	1	MM5 Area of a triangle	MG2H-4
23	1	FsDr3 BAC	MGP-3
24	1	AM3 Linear Equation	MG2H-3, 9
25	1	FsDr1 Car Insurance	MGP-6

Section II

Question	Marks	Content	Syllabus outcomes
26 (a)	2	DS6 Capture recapture	MG2H-2
26 (b)	2	FSHe2 Medication	MG2H-5
26 (c)	2	MM6 Simple Distance	MG2H-4
26 (d)	2	FM2 Appreciation	MGP-6
26 (e) (i)	1	PB1 Relative frequencies	MGP-8
26 (e) (ii)	1	PB1 Relative frequencies	MGP-8
26 (f)	2	MM4 SA Sphere	MG2H-4
26 (g)	3	FsCo1 Mobile Phone	MGP-6
27 (a)	2	MM3 Similar Triangles	MGP-4
27 (b)	2	FsHe2 Medication	MG2H-5
27 (c) (i)	2	AM4 Simultaneous equations	MG2H-3
27 (c) (ii)	2	AM4 Simultaneous equations	MG2H-3
27 (d) (i)	3	DS4 Outliers	MG2H-2 MG2H-7
27 (d) (ii)	1	DS4 Standard Deviation	MGP-10
27 (e)	3	FsCo2 Rates (File downloads)	MGP-5
28 (a)	1	MM4 Annulus Area	MG2H-4
28 (b)	2	DS5 z-scores	MG2H-2 MG2H-7
28 (c)	3	FSRe2 Simpson's rule – Volume	MG2H-4
28 (d)	2	AM3 Sub in and solve (Temp.)	MG2H-3
28 (e) (i)	2	FsHe1 Scatterplot and line of best fit	MG2H-7
28 (e) (ii)	1	FsHe1 Scatterplot and line of best fit	MG2H-2
28 (e) (iii)	1	FsHe1 Correlation	MG2H-7
28 (f) (i)	1	AM4 Breakeven using Quadratic	MG2H-3
28 (f) (ii)	2	AM4 Breakeven using Quadratic	MG2H-3
29 (a)	2	FM4 Credit Card	MG2H-6
29 (b)	2	FM5 Home loan graphs	MG2H-6
29 (c) (i)	2	MM2, FsRe2 Scale Drawing roof area	MG2H-9 MG2H-4
29 (c) (ii)	3	MM2, FsRe2 Volume	MG2H-4
29 (d) (i)	1	DS4 Grouped data	MG2H-2 MG2H-10
29 (d) (ii)	3	DS4 Grouped data	MG2H-2 MG2H-10

Question	Marks	Content	Syllabus outcomes
29 (e)	2	AM5 Non Linear Modelling	MG2H-3
30 (a)	2	FsRe3 Energy Use	MG2H-5
30 (b)	3	PB2 Probability Tree	MG2H-8
30 (c) (i)	1	FM5 Present Value Table	MG2H-6
30 (c) (ii)	2	FM5 Present Value Table	MG2H-6
30 (d)	2	FsDR3 Stopping Distances	MGP-5
30 (e) (i)	1	MM5 Trig, diagram given	MG2H-4
30 (e) (ii)	4	MM5 Trig, diagram given	MG2H-4